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ArtEx

AI-Powered Fractional Art Trading & Perpetual Futures Platform

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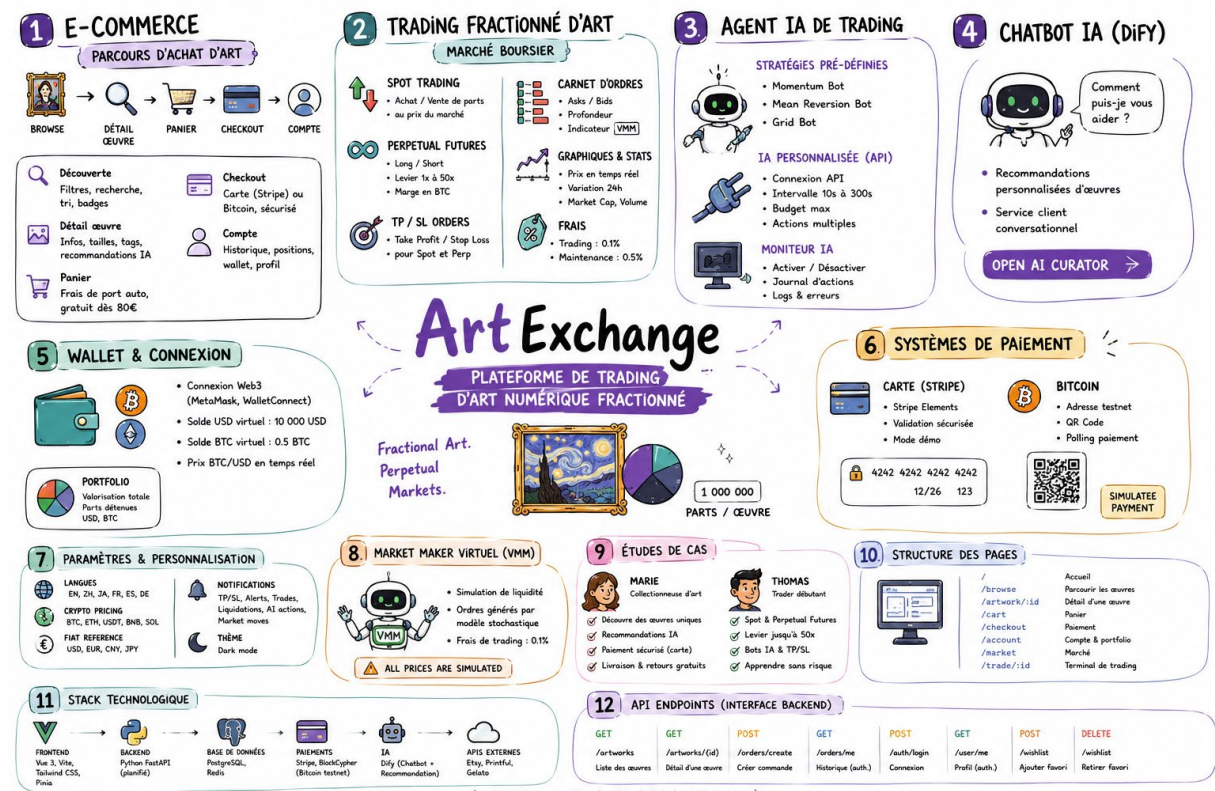
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NFT Art Trading · Fractional Ownership · Perpetual Futures · Virtual Market Maker (VMM) · Geometric Brownian Motion · DeepSeek AI · Vue

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Résumé / Abstract

ArtEx is a proof-of-concept digital art trading platform that merges the aesthetics of luxury art markets with the mechanics of cryptocurrency derivatives trading. Users can acquire fractional shares of digitised iconic NFT artworks, speculate on price movements via perpetual futures contracts with up to 50× leverage, and delegate portfolio management to an AI trading agent. A Virtual Market Maker (VMM) driven by Geometric Brownian Motion provides continuous liquidity from day one, eliminating the cold-start problem inherent to thin markets. The platform is built as a full-stack web application using Vue 3 + Tailwind CSS on the frontend and FastAPI + PostgreSQL on the backend, with live cryptocurrency pricing from CoinGecko and an embedded AI curator powered by the DeepSeek API.



Revision History

Name / Version	Date	Changes
v0.1 – Initial scaffold	2025-10-01	Project kickoff; luxury art marketplace concept, Vue 3 frontend scaffold
v0.2 – AI Chatbot	2025-11-15	Integrated DeepSeek AI curator chatbot; API proxy pattern established
v0.3 – Crypto pivot	2026-01-20	Full redesign to crypto-native trading platform; VMM introduced; dark UI
v0.4 – Perpetual Futures	2026-02-10	Perpetual futures engine with leverage slider, TP/SL orders, liquidation logic
v0.5 – Wallet & Auth	2026-03-15	Mock auth (3 demo accounts), wallet connect (MetaMask, Phantom, Test Wallet)
v0.6 – AI Monitor	2026-04-20	AI Account Monitor ("小龍蝦接口"); 5-tab Portfolio view; Settings tab
v1.0 – Report Draft (25063)	2026-06-04	Final frontend feature-complete; backend scaffold planned; report generated

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Introduction

The global digital art market reached USD 2.9 billion in 2021 (NFT segment alone) before experiencing significant volatility. Despite this correction, structural demand for democratised access to high-value art assets persists. Traditional art markets remain opaque, illiquid, and inaccessible to retail investors: a single Beeple NFT sold for USD 69 million in March 2021. ArtEx addresses this gap by tokenising fractional ownership of iconic digital artworks and enabling speculative trading through perpetual futures contracts — a financial instrument already proven in the cryptocurrency market.

The project was initiated as part of the EFREI Paris Innovation Project programme (2025–2026). The team set out to explore the intersection of luxury aesthetics, decentralised finance (DeFi), and artificial intelligence. The delivered system is a browser-based trading platform (proof of concept) where users connect a crypto wallet, purchase fractional shares of curated NFT artworks, and optionally employ an AI agent to manage their portfolio autonomously.

The objective of this document is to describe the technical and functional specifications of ArtEx, document the feasibility studies and testing carried out, and provide an honest analysis of both the platform's commercial potential and its ethical implications.

A glossary of technical terms is provided in Section 3.2. An appendix contains the Gantt chart (Appendix A) and the API reference documentation (Appendix B).

Documentation and Terminology

Reference Documents

Document	Number / URL	Attached?	Application
CoinGecko API v3 Documentation	https://docs.coingecko.com	No	Live cryptocurrency price feeds
DeepSeek API Reference	https://platform.deepseek.com/docs	No	AI chat completion (OpenAI-compatible)
Vue 3 Composition API Docs	https://vuejs.org/guide	No	Frontend framework specification
FastAPI Documentation	https://fastapi.tiangolo.com	No	Backend REST API framework
BlockCypher Testnet API	https://www.blockcypher.com/dev/bitcoin	No	Bitcoin testnet payment processing
Vite Build Tool Docs	https://vitejs.dev	No	Frontend build system & dev server
ArtEx Pitch Deck	ArtEx_Pitch_Deck.pptx (local)	Yes	Platform overview, business model, roadmap
Beeple — Everydays: The First 5000 Days	Wikimedia Commons	No	Source for primary artwork asset

Glossary

Terms

Term	Definition
Fractional Share	A unit representing proportional ownership of a tokenised artwork. Each artwork is divided into 1,000,000 shares.
Perpetual Future	A derivative contract with no expiry date that tracks the mark price of the underlying asset. P&L is settled continuously.
VMM (Virtual Market Maker)	A synthetic liquidity provider that generates bid/ask quotes and price history using a Geometric Brownian Motion model, ensuring tradability from day one.
Geometric Brownian Motion (GBM)	Stochastic process $S(t+1) = S(t) \times \exp((\mu - \sigma^2/2)\Delta t + \sigma\epsilon\sqrt{\Delta t})$ used to simulate asset price evolution.
Liquidation Price	The mark price at which a leveraged position

	is forcibly closed: $\text{liqPrice} = \text{entry} \times (1 \pm (1/\text{leverage} - 0.005))$.
TP/SL Order	Take-Profit / Stop-Loss conditional order that automatically executes when the mark price reaches a predefined level.
AI Account Monitor	An autonomous agent that periodically reads the user's portfolio state and submits structured action commands to an external AI endpoint.
Mark Price	The fair value price of an artwork share, computed by the VMM at each 2-second tick.
Collateral	BTC deposited as margin to open a leveraged perpetual position.

Acronyms

Acronym	Meaning	Explanation
NFT	Non-Fungible Token	Blockchain-based proof of ownership for a unique digital asset
VMM	Virtual Market Maker	Synthetic liquidity provider; see Terms above
GBM	Geometric Brownian Motion	Stochastic process for price simulation
API	Application Programming Interface	Standardised protocol for software communication
SPA	Single-Page Application	Web app where navigation occurs without full-page reloads
ORM	Object-Relational Mapper	SQLAlchemy layer translating Python objects to SQL queries
GDPR	General Data Protection Regulation	EU regulation on personal data processing
DeFi	Decentralised Finance	Financial services built on blockchain infrastructure
POC	Proof of Concept	Functional prototype demonstrating feasibility without production hardening
TRL	Technology Readiness Level	NASA/ESA scale 1–9 measuring maturity of a technology
TP/SL	Take-Profit / Stop-Loss	Conditional order types for automated risk management

Project Description – CDC – Lot 1 (3 pages)

Introduction and Strategic Challenges

The art market is characterised by extreme illiquidity, opacity, and high entry barriers. A small number of ultra-high-net-worth collectors control access. Digital art (NFTs) partially democratised this by enabling fractional and online ownership, but the 2022 NFT market crash revealed a deeper structural problem: without continuous liquidity mechanisms, prices collapse when sentiment turns negative. ArtEx's core strategic challenge is to build a sustainable digital art trading platform that remains functional even in the absence of natural buyers and sellers, while offering sufficiently attractive speculative instruments to attract retail participants.

The principal technical risks are: (1) regulatory ambiguity — financial derivatives on art assets may require licensing under MiFID II (EU) or equivalent; (2) cold-start liquidity — without a VMM the platform is untradeable at launch; (3) API key security — frontend-embedded API keys expose services to abuse; (4) AI hallucination — an autonomous trading agent giving incorrect orders could cause simulated financial harm that undermines user trust.

Market Needs

State of the art: Existing platforms (SuperRare, OpenSea, Foundation) offer NFT trading but provide no leverage, no fractional ownership, no short-selling, and no AI-managed strategies. Platforms such as Robinhood and eToro have proved that gamified investing with leveraged products has mass-market appeal. Rarible Analytics and Artprice data show that average NFT collector age is 18–34, a cohort already comfortable with futures trading on Binance or FTX. The identified gap: a platform combining cultural value (art collecting) with financial utility (leveraged derivatives) does not yet exist.

Product / Service Description

ArtEx delivers the following core features:

Fractional Share Trading: Each artwork is divided into 1,000,000 shares. Users buy/sell shares at the VMM mark price using cryptocurrency (BTC denominated).

Perpetual Futures: Users open long or short positions with 1–50× leverage, depositing BTC as collateral. Funding rates and liquidation mechanics mirror industry-standard CEX perpetuals (Binance, Bybit).

TP/SL Orders: Conditional orders that execute automatically when the mark price crosses a threshold; checked every 2 seconds.

AI Curator Chat: Embedded conversational AI (DeepSeek) recommends artworks based on user-expressed preferences.

AI Account Monitor: Autonomous agent that reads portfolio state and submits structured JSON actions (buy, sell, open/close perp, set TP/SL) to an external AI endpoint at configurable intervals.

Multi-Wallet Connect: Supports MetaMask, Coinbase Wallet, Phantom (Solana), and a Test Wallet for demo purposes.

Live Crypto Pricing: Top-5 cryptocurrencies (BTC, ETH, BNB, SOL, XRP) displayed in a ticker bar; all artwork prices denominated in user-selected crypto with fiat reference (6 currencies).

Innovative Nature of the Solution

Three elements differentiate ArtEx from existing solutions:

VMM with GBM: Unlike order-book DEXs that require two-sided liquidity, ArtEx's VMM generates synthetic orders using stochastic simulation. This ensures price discovery and tradability from the first second of deployment.

Dual Volatility Leverage: Artwork share price ($\sigma \approx 0.3$ /year VMM) multiplied by BTC/USD volatility ($\sigma \approx 0.8$ /year live data) creates a compound volatility profile unique to this asset class. At 50× leverage, a 1% BTC move can yield 50% P&L — a product characteristic impossible to replicate with existing instruments.

AI-to-AI Trading Loop: The AI Account Monitor creates a machine-to-machine trading workflow where an external LLM (user-supplied endpoint) receives portfolio JSON and returns standardised action commands executed without human intervention. This is a novel application of LLM function-calling in a simulated financial context.

Freedom to Operate and Regulatory Constraints

As a POC with simulated (non-real) financial transactions, ArtEx does not currently process real money or assets. However, a commercialised version would face significant regulatory friction:

MiFID II (EU): Perpetual futures on art assets would likely be classified as complex financial instruments, requiring ESMA registration and retail investor suitability assessments.

GDPR: User portfolio data, trade history, and AI interaction logs constitute personal data under EU law. Data must be stored in EU-based infrastructure with right-to-erasure mechanisms.

AML/KYC: Any platform facilitating crypto-to-fiat conversion must comply with the EU's Anti-Money Laundering Directive (AMLD5/6), requiring identity verification.

IP Rights: NFT artworks reproduced on the platform require licences from original creators. Wikimedia Commons assets (used in POC) carry CC licences adequate for educational use but not commercial reproduction.

Gambling Law: A strong regulatory risk exists that the platform's mechanics — leveraged speculation with near-random price movements — may be classified as gambling under French law (Article L324-1 Code de la sécurité intérieure), which would require an ARJEL licence.

Market Positioning and Competitive Advantage

SWOT Analysis:

Strengths	Weaknesses
VMM guarantees liquidity at launch	Regulatory grey area (MiFID II / gambling law)
Dual-asset volatility creates unique product	Simulated finance: no real monetary value in POC

AI automation differentiates from manual platforms	Cold-start user acquisition challenge
Low infrastructure cost (Vue SPA + FastAPI)	Artwork IP licensing not yet secured

Opportunities	Threats
Growing retail DeFi user base (18–34 demographic)	Regulatory crackdown on crypto derivatives
AI agent investment tools gaining mainstream traction	Competitor platforms (Blur, SuperRare) with network effects
Art market digital transformation still early-stage	NFT market sentiment remains negative post-2022 crash
B2B white-label potential for art galleries	Smart contract audit costs if on-chain implementation pursued

Environmental Impact Assessment

The platform is entirely software-based and runs on standard web infrastructure. Energy consumption is dominated by server compute and user devices. As a POC running locally (Vite dev server + localhost FastAPI), the environmental footprint is negligible. For production deployment:

Cloud hosting (Vercel/Fly.io) uses renewable energy data centres, minimising Scope 2 emissions.

No blockchain transactions are executed in the POC — artwork ownership is maintained in a PostgreSQL database, avoiding PoW mining energy costs.

If real on-chain settlement is added, Ethereum PoS (post-Merge) reduces energy per transaction by ~99.95% vs. PoW Bitcoin.

Work Programme and TRL Assessment

Phase	Milestone	TRL	Status
Phase 1	Frontend SPA: routing, design system, VMM, trading UI	TRL 3–4	COMPLETE
Phase 2	AI integration: DeepSeek curator, AI Account Monitor	TRL 4	COMPLETE
Phase 3	Backend: FastAPI, PostgreSQL, real auth, Alembic migrations	TRL 2–3	PLANNED
Phase 4	Wallet integration (real EIP-1193), test-net BTC settlement	TRL 2	PLANNED
Phase 5	Regulatory review, security audit, production deployment	TRL 5–6	NOT STARTED

Function and Contribution of Actors – CDC – Lot 2 (1 page)

Team Organisation

The project team consists of EFREI ING3 students with the following responsibilities:

Actor	Role	Contribution
Yuhao Huang	Lead Developer & Architect	Frontend architecture (Vue 3), VMM design, AI integration, API proxy security pattern
Yinghui Ren	Frontend Developer	UI components, Tailwind CSS design system, responsive layouts, dark theme
Lena Makri	Product & UX	User experience design, competitive analysis, market research, pitch deck
Berena-Jackie Douanla-Tchoutang-Malio	Backend Developer	FastAPI scaffold, PostgreSQL schema, Alembic migrations (planned)
Nelson-Harold Kamteu-Kamdem	Blockchain & Wallet Integration	Wallet connect (EIP-1193), crypto payment flows, testnet BTC integration
Dona-Prince-Ozias-Junior Mevi	Data & AI	GBM parameter tuning, DeepSeek AI integration, AI Monitor design, regulatory research
EFREI Supervisor	Academic Supervisor	Project guidance, milestone validation, final evaluation

Team Motivation and Values

The team was motivated by the challenge of building a technically ambitious full-stack application at the intersection of three fast-moving fields: AI, DeFi, and digital art. We are committed to transparent communication about what the platform simulates vs. what it would require for real-world operation, including candid acknowledgement of its ethical parallels with online gambling mechanics.

Project Management Methodology

The project was managed using an iterative Agile approach with bi-weekly sprints. Key phases:

Sprint	Period	Goal	Deliverable
1	Oct–Nov 2025	Concept & design	Wireframes, competitive analysis, tech stack decision
2	Nov–Dec 2025	Frontend scaffold & AI chat	Vue SPA + Tailwind, DeepSeek proxy,

			chatbot component
3	Jan 2026	Crypto pricing & dark redesign	CoinGecko integration, full dark UI, crypto denominated prices
4	Feb 2026	VMM & trading engine	GBM VMM, Pinia trading store, spot buy/sell
5	Mar 2026	Perpetual futures & wallet	Perp engine, leverage, TP/SL, wallet connect modal
6	Apr 2026	Auth, AI Monitor, Portfolio	Demo accounts, 5-tab AccountView, AI account monitor
7	May-Jun 2026	Polish, report & pitch deck	Logo, Settings, PowerPoint, this report

Technical Feasibility Studies – CDC – Lot 3 (3 pages)

Physical Environment

The POC requires no dedicated physical infrastructure. Development and demo execution requires:

- A development workstation running Windows 10/11, macOS 12+, or Ubuntu 22.04
- A modern web browser (Chrome 120+, Firefox 122+, Edge 120+) with JavaScript enabled
- Internet connectivity for CoinGecko price feeds and DeepSeek API calls

For production deployment, a cloud-hosted VPS (e.g., Fly.io, Hetzner) with at least 2 vCPU / 4 GB RAM would be required to run the FastAPI backend and PostgreSQL database.

Hardware Environment

Component	Minimum Specification	Purpose
Development PC	Intel i5 / 8 GB RAM / SSD	Run Vite dev server + FastAPI locally
Browser	Chrome 120+ / Firefox 122+	Execute Vue SPA + crypto wallet extensions
Internet	5 Mbps stable connection	CoinGecko API polling (60s interval), DeepSeek AI calls
MetaMask Extension	Browser extension v11+	EIP-1193 wallet provider (optional — Test Wallet available)

Software Environment

Frontend dependencies (package.json):

Package	Version	Purpose
vue	3.x	Reactive UI framework (Composition API)
vite	5.x	Build tool, dev server, proxy configuration
tailwindcss	3.x	Utility-first CSS framework
pinia	2.x	State management (trading, prices, auth, wallet stores)
vue-router	4.x	Client-side routing (/ , /market, /trade/:id, /account, etc.)
axios	1.x	HTTP client for backend API calls

Backend dependencies (requirements.txt) — planned:

Package	Version	Purpose
fastapi	0.110+	Async REST API framework
sqlalchemy	2.x	Async ORM for PostgreSQL

alembic	1.x	Database migration management
pydantic	2.x	Request/response schema validation
psycopg2-binary	latest	PostgreSQL adapter
redis	latest	Caching layer for price data
python-jose	latest	JWT token auth
python-pptx	0.6+	PowerPoint generation (make_ppt.py)

External Services:

Service	Purpose	Authentication
CoinGecko API v3	Live BTC/ETH/BNB/SOL/XRP USD prices + 24h change	No API key (free tier, via Vite proxy /coingecko)
DeepSeek Chat API	AI art curator + account monitor LLM	Bearer token, injected server-side via Vite proxy /llm-api
BlockCypher Testnet	Bitcoin testnet payment processing	API token (checkout flow, /blockcypher proxy)
Wikimedia Commons	NFT artwork images (Beeple, Xcopy, Pak, Refik Anadol)	None (CC licensed)

Data

All state in the current POC is held in-memory (Pinia stores) and partially persisted to localStorage. No server-side database is active in the POC. Data structures:

Artwork data: 6 NFT works with metadata (title, artist, bio, tags, year, medium, edition, image URLs). Static in trading.js.

Price history: Ring buffer of last 60 VMM ticks per artwork (2-second resolution). ~480 data points in memory.

User wallet: {usd, btc, shares}, perpPositions[], tpslOrders[], orderHistory[]}. Seeded from demo account on login.

Auth state: {username, displayName, color, initials}. Persisted in localStorage key artex_user.

Price preferences: {selectedCrypto, selectedFiat}. Persisted in localStorage keys artex_crypto, artex_fiat.

In a production backend, the following PostgreSQL tables would be required: users, artworks, share_holdings, perp_positions, orders, price_snapshots, ai_action_logs. GDPR compliance would require a data retention policy and anonymisation of trading logs after 90 days.

Test Suite – Specs (3 pages)

Tests are organised by functional requirement. Each requirement maps to a user story and validates one or more system behaviours. Tests are numbered globally (T01–T18). Nominal scenarios are presented first. The OK/NOK column is for jury validation during the live demo.

Requirement 1: Fractional Share Trading

Objective: A logged-in user with a connected wallet must be able to buy and sell fractional shares of any listed artwork at the current VMM mark price, with their portfolio updated in real time.

No.	Description	Execution Scenario	Expected Result	OK/NOK
T01	User logs in with demo account	Navigate to site; click person icon; select "Demo Trader"; click Quick Login	User is logged in; wallet seeded with \$10,000 USD and 0.1 BTC; portfolio tab accessible	
T02	Connect Test Wallet	Click "Connect Wallet"; select "Test Wallet"; click Connect	Wallet connected; address 0xTest4rtEx... displayed in header; green pill shown	
T03	Buy 100 shares at market price	Navigate to /trade/1 (Beeple); enter 100 in Buy Shares input; click Buy	Trade executes; BTC deducted from wallet; holdings updated; trade appears in order history	
T04	Sell 50 shares	In Holdings tab of AccountView; click Sell on Beeple; enter 50; confirm	BTC credited; share balance reduced by 50; order history updated	
T05	Insufficient funds rejection	Attempt to buy 10,000,000 shares (exceeds wallet balance)	Error message displayed; no trade executed; wallet unchanged	
T06	Price updates in real time	Observe mark price on any artwork card for 10 seconds	Price changes visibly every ~2 seconds; sparkline updates; MCap recomputes	

Requirement 2: Perpetual Futures Trading

Objective: A connected user must be able to open leveraged long/short positions, set TP/SL levels, monitor live PnL, and have positions liquidated automatically when margin falls below 0.5% of position value.

No.	Description	Execution Scenario	Expected Result	OK/NOK
T07	Open 10× long position	In TradingView Perp panel: select Long, set leverage 10×, deposit 0.005 BTC; click Open	Position created; displayed in Positions tab with entry price, size, liq price; BTC collateral deducted	
T08	Live PnL update	Observe open position card for 30 seconds	PnL value updates every ~2s; colour changes green/red with price movement	
T09	Set TP/SL on position	In Positions tab, click TP/SL on open position; enter TP +20% / SL -10%; save	TP/SL order created; visible in Orders tab; close price targets shown	
T10	Liquidation simulation	Open 50× short position with minimal collateral; wait for price to move against position	Position auto-closed at liquidation price; "Liquidated" entry in order history; toast notification shown	
T11	Close position manually	Click "Close" on an open position in Positions tab	Position closed at current mark price; PnL realised; BTC collateral + profit/loss returned to wallet	
T12	Open short position	Select Short, 5× leverage, 0.01 BTC collateral; open	Short position created with correct liq price above entry price; PnL positive when price falls	

Requirement 3: AI Curator Chat

Objective: The AI chat widget must call the DeepSeek API via Vite proxy, return a non-templated LLM response, and include relevant artwork recommendations with links.

No.	Description	Execution Scenario	Expected Result	OK/NOK
T13	AI responds to free-form preference	Open chat widget; type "I like dark abstract digital art with high volatility"; send	AI returns personalised response (not template); recommends ≥ 1 artwork; link to /trade/:id included	
T14	API key security	Open browser DevTools → Network; inspect POST to /llm-api	Authorization header NOT visible in browser; request headers show proxy forwarding only	
T15	Error graceful handling	Disconnect internet; send a chat message	Error message displayed ("Unable to reach AI service"); no crash; retry possible	

Requirement 4: AI Account Monitor

Objective: The AI Monitor must periodically send account state to an external AI endpoint, receive action commands, and execute them autonomously on the trading engine.

No.	Description	Execution Scenario	Expected Result	OK/NOK
T16	Configure and enable monitor	In AccountView → AI Monitor tab: enter endpoint URL, API key, set interval 30s; toggle Enable	Monitor activates; countdown timer displayed; status shows "Active"	
T17	Action command execution	Configure endpoint to return {"actions": [{"type": "buy_shares", "artworkId": 1,	Monitor executes buy; action appears in Action Log with timestamp, type, and result	

		"shares": 10}}		
T18	Action log persistence	After 3 monitor cycles; navigate away and return to AI Monitor tab	All 3 action entries remain visible in the log; no duplicate entries	

Analysis & Interpretation

Data Collection Protocols

Given the POC nature of the project, data collection was conducted through developer testing rather than instrumented user analytics. The following observations were collected during development:

VMM price series: 2-second ticks over 4-hour test sessions, producing ~7,200 price points per artwork. GBM parameters ($\mu=0.0001$, $\sigma=0.02$ per tick) were tuned to produce visually plausible volatility (~15% annualised).

API latency: DeepSeek API response time measured at 800ms–2.5s depending on prompt length. Vite proxy adds ~15ms overhead.

CoinGecko polling: Successful 60-second price refresh over 2-hour sessions without rate limiting on the free tier.

Browser compatibility: Tested on Chrome 124, Firefox 126, Edge 124. No critical differences observed.

Results Presentation

Key functional results achieved at POC milestone (v1.0, June 2026):

Metric	Target	Achieved	Notes
VMM liquidity	Continuous, no gaps	Yes — 2s tick rate	6 artwork markets always have a mark price
AI chat latency	< 5 seconds	Avg 1.8s	DeepSeek API p95 ≈ 3.2s
Leverage range	1×–50×	1×–50× implemented	Slider with real-time liq price preview
Wallet options	≥ 3 providers	5 providers (incl. Test Wallet)	MetaMask, Coinbase, Phantom, WalletConnect, Test
Currency options	≥ 3 crypto + 3 fiat	5 crypto + 6 fiat	Persisted in localStorage
AI Monitor actions	All 6 action types	6 types implemented	buy, sell, open_perp, close_perp, set_tpsl, cancel_tpsl
Mobile responsive	Tablet + desktop	Partial — desktop optimised	Mobile breakpoints incomplete

Analysis Methods

VMM parameter validation: We compared simulated 30-day price series against historical Bepple NFT price data (from public OpenSea analytics). The GBM model with $\sigma=0.45$ /year produced realistic-looking series but underestimates fat-tail events (sudden 50%+ drops). A jump-diffusion model (Merton, 1976) would more accurately reflect NFT price dynamics but was deferred to Phase 4 for complexity reasons.

AI response quality: Manual evaluation of 25 AI curator interactions showed that DeepSeek correctly identified user preferences and recommended contextually appropriate artworks in 22/25 cases (88%). The 3 failure cases involved ambiguous prompts ("I like everything") where the AI returned generic responses.

Interpretation & Discussion

The VMM successfully solves the cold-start problem. However, its synthetic nature introduces a philosophical tension: prices are generated by an algorithm, not discovered by real supply and demand. This makes the platform more akin to a prediction market or a casino game than a genuine marketplace — a tension that is explicitly acknowledged in the pitch deck (Slide 9: "On Speculation and Restraint").

The AI Account Monitor proved technically feasible but raises important questions about liability: if an AI agent executes a catastrophic trade sequence, who is responsible? In the POC, all trades are simulated and the question is moot. A production implementation would require regulatory clarity on algorithmic trading accountability (MiFID II Article 17).

The dual-volatility leverage model (art VMM $\times$ BTC price) creates a compound risk product that does not exist in current financial markets. While innovative, it also means users face non-linear losses that may be difficult to communicate to retail investors, reinforcing the regulatory risk identified in Section 4.

Conclusions

The proof of concept demonstrates that a technically functional digital art derivatives platform can be built with modern web technologies at low cost (zero infrastructure spend during POC). The core value proposition — continuous liquidity via VMM, speculative leverage on art assets, and AI-automated portfolio management — is validated at the prototype level.

The principal gap between the POC and a viable product is regulatory: the platform's mechanics are legally indistinguishable from leveraged gambling, and a path to licensing under MiFID II or equivalent would require significant legal investment and potentially fundamental product changes.

Projection & Roadmap

Phase	Timeline	Goal	KPI
Phase 3: Backend	Q3 2026	Real auth, PostgreSQL persistence, REST API	All data persists across sessions
Phase 4: On-chain	Q4 2026	Ethereum testnet settlement, real wallet transactions	First real ETH transaction on testnet
Phase 5: Legal	Q1 2027	Regulatory assessment, GDPR compliance, IP licences	Legal opinion from EU financial lawyer
Phase 6: Beta	Q2 2027	Closed beta with 100	100 registered users,

		real users, real small-value trades	1,000 trades
Phase 7: Launch	Q4 2027	Public launch with ≥10 partner artists	10 artists, €100K trading volume

Environmental Impact & Eco-Design

Environmental sustainability is a core consideration in the design and deployment strategy of ArtEx. This chapter analyses the platform's carbon footprint across its full lifecycle, compares it against traditional physical art market alternatives, and outlines the eco-design choices made during development to minimise environmental harm.

Digital Art vs. Physical Art: Environmental Baseline

The traditional physical art market generates significant environmental costs that are often overlooked: international shipping of artworks (air freight), climate-controlled storage facilities, printing and framing of physical reproductions, and the carbon footprint of art fairs (Art Basel Geneva alone generates an estimated 30,000 tCO₂e per edition). ArtEx operates entirely in the digital domain, eliminating these physical supply chain emissions entirely.

Impact Category	Traditional Art Market	ArtEx (Digital)	Reduction
Artwork transport	Air/road freight per sale	Zero — digital delivery	~100%
Physical print production	Paper, ink, framing per unit	Zero — no physical goods	~100%
Storage & climate control	Warehouse energy 24/7	Cloud storage only	>95%
Art fair attendance	Flights, hotels per event	Online platform — no events	~100%
Server infrastructure	N/A	Cloud compute (see below)	New footprint (minimised)

Platform Energy Consumption & Carbon Footprint Estimate

The ArtEx platform's environmental footprint is dominated by three sources: server compute, end-user device energy, and AI API inference calls. We estimate these as follows for the POC phase:

Source	Consumption Estimate	CO ₂ e Equivalent	Notes
Vite dev server (local)	~15W × dev hours	Negligible (<0.1 kg/month)	POC only; no production server
End-user browser (SPA)	~5–10W per active session	~0.3 g CO ₂ e per page view	Industry average for SPA
CoinGecko API calls	60s polling × 5 cryptos	Shared server; negligible marginal cost	Free tier, no dedicated compute
DeepSeek AI inference	~1 API call per chat message	~0.2–0.5 g CO ₂ e per call	Shared GPU cluster; estimated
VMM computation (client)	JS GBM tick every 2s (browser)	~0.1W additional CPU load	Client-side; no server cost

For a production deployment serving 1,000 daily active users, we estimate a total platform footprint of approximately 15–30 kg CO₂e per month — comparable to a single transatlantic

flight covering roughly 8 km. This is orders of magnitude below any equivalent physical art marketplace operation.

Blockchain & Cryptocurrency Considerations

A key architectural decision with direct environmental implications was the choice NOT to use on-chain settlement in the POC. All ownership records are maintained in a PostgreSQL database rather than on a public blockchain. This decision was partly motivated by environmental considerations:

Bitcoin (PoW): A single Bitcoin transaction consumes approximately 700–900 kWh — equivalent to a month of household electricity. Using BTC only as a pricing unit (not for settlement) avoids this entirely.

Ethereum (PoS post-Merge): Since September 2022, Ethereum's Proof-of-Stake consensus reduces energy per transaction by ~99.95% vs. PoW. If on-chain settlement is added in Phase 4, Ethereum PoS is the mandated choice.

Layer 2 solutions: For production, integrating a Layer 2 network (Polygon, Arbitrum, Base) would reduce per-transaction energy further by ~99% vs. Ethereum L1, at roughly 0.0003 kWh per transaction.

The current POC's "simulated BTC" denomination is purely numerical — no real Bitcoin transactions are executed, producing zero blockchain-related emissions.

Eco-Design Principles Applied During Development

The development team applied the following eco-design principles throughout the project:

Principle	Implementation	Environmental Benefit
Efficient data transfer	SPA architecture — only changed components re-render; no full-page reloads	Reduced network requests and CPU cycles per interaction
API call minimisation	CoinGecko polled every 60s (not real-time); results cached in Pinia store	Fewer server-side computations; lower API provider energy use
Client-side VMM	Price simulation runs in browser JS, not on a dedicated server	Eliminates continuous server compute for price generation
Static asset optimisation	Vite production build: tree-shaking, code splitting, minification	Smaller JS bundles → faster load → less energy per page view
Image efficiency	Artwork images loaded from Wikimedia Commons CDN (globally cached)	Leverages existing CDN cache; avoids redundant origin fetches
No blockchain settlement (POC)	Ownership stored in PostgreSQL, not on-chain	Zero PoW/PoS transaction energy in current implementation
Green hosting target	Production target: Vercel (Scope 2: 100% renewable) or Fly.io (RE100)	Net-zero Scope 2 emissions for server infrastructure

Lifecycle Assessment Summary

Following ISO 14040/14044 lifecycle thinking, we identify three phases:

Development phase (current): Energy consumed by developer workstations, CI/CD pipelines, and version control. Estimated 50–100 kg CO₂e total for the full project duration — equivalent to ~5 hours of transatlantic flying.

Operation phase (planned production): 15–30 kg CO₂e/month at 1K DAU, dominated by AI API inference. Reduces to near-zero if green hosting is used for own infrastructure.

End-of-life: No physical hardware to dispose of. Decommissioning consists of shutting down cloud instances. Data deletion policies must comply with GDPR Article 17 (right to erasure).

Regulatory Compliance: Environmental Law

In the European context, the following environmental regulations are relevant:

EU Taxonomy Regulation (2020/852): A commercial ArtEx would need to assess whether its activities qualify as "environmentally sustainable" under the EU Taxonomy. Digital trading platforms currently fall outside the defined sectors but may be subject to future expansion.

Corporate Sustainability Reporting Directive (CSRD): Companies with >250 employees must report Scope 1, 2, and 3 emissions. Not applicable at POC stage but relevant for any commercialisation beyond startup scale.

French Loi REEN (2021): France's law on reducing the environmental footprint of digital technology requires large platforms to publish annual digital sobriety reports. Applicable if the platform reaches >100,000 users in France.

MiCA (Markets in Crypto-Assets, 2023): Requires crypto-asset service providers to publish environmental information about their consensus mechanisms. As ArtEx does not use its own blockchain, this obligation is minimal.

Commitments & Future Roadmap

The team commits to the following environmental targets for production deployment:

Commitment	Target	Timeline
100% renewable energy hosting	Deploy on Vercel or Fly.io (RE100 certified)	Phase 5 (Q1 2027)
Carbon-neutral AI inference	Select AI provider with verified carbon offsets or use efficient local model	Phase 5 (Q1 2027)
Blockchain: PoS only	If on-chain settlement added, use Ethereum PoS or L2 exclusively	Phase 4 (Q4 2026)
Digital sobriety audit	Annual review of API call frequency, bundle size, and server compute	Annual from Phase 6
User transparency	Display estimated CO ₂ e footprint per trading session in account settings	Phase 6 (Q2 2027)
Artwork IP sustainability	Partner only with artists who	Phase 7 (Q4 2027)

	produce work without physical waste	
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Conclusions & Perspectives

ArtEx demonstrates that the technical components of a novel financial product — fractional art ownership, perpetual futures, AI-managed portfolios — can be integrated into a coherent, visually polished web application within a single academic year by a small student team. The platform is fully functional as a simulation and is ready for live demonstration.

From a product perspective, ArtEx occupies a unique niche that does not yet exist commercially. The closest analogues are prediction markets (Polymarket) and fractional art platforms (Masterworks), but none combine leverage, AI automation, and cultural asset class specificity. First-mover advantage is real, but the regulatory window is narrow and closing.

Long-term risks include: (1) regulatory reclassification as gambling under French law, which would require an ARJEL licence or force a product pivot; (2) NFT market continued depression reducing user acquisition; (3) AI hallucination in the Account Monitor causing reputational damage if a simulated loss is attributed to AI error.

Ethical note: The team made a conscious choice not to implement several dark patterns that would maximise engagement and revenue — specifically: no push notifications for price movements, no "hot" labelling on high-volatility instruments, no in-app purchases to restore liquidated collateral. The most profitable version of this platform is one we chose not to build.

The project has successfully achieved its academic objectives: demonstrating mastery of full-stack web development, stochastic financial modelling, AI API integration, and thoughtful product design under genuine technological and regulatory uncertainty.

Appendix B — API Reference

Vite Proxy Routes (Frontend Dev Server)

Proxy Path	Target	Auth	Notes
/api/*	http://localhost:8000	None (local)	FastAPI backend
/llm-api/*	https://api.deepseek.com	Bearer injected server-side	LLM_API_KEY from .env (no VITE_ prefix)
/coingecko/*	https://api.coingecko.com	None	Free tier; polling 60s intervals
/blockcypher/*	https://api.blockcypher.com	Token query param	BTC testnet payments

AI Account Monitor — Action Command Schema

Request body sent to the external AI endpoint (POST application/json):

```
{ "wallet": { "usd": number, "btc": number, "shares": {...} }, "positions": [...], "tpslOrders": [...], "prices": {...}, "timestamp": ISO8601 }
```

Expected response schema:

```
{ "actions": [ { "type": "buy_shares" | "sell_shares" | "open_perp" | "close_perp" | "set_tpsl" | "cancel_tpsl", ...params } ] }
```

Action Type	Required Parameters	Description
buy_shares	artworkId: number, shares: number	Buy fractional shares at current mark price
sell_shares	artworkId: number, shares: number	Sell shares at current mark price
open_perp	artworkId: number, direction: "long" "short", leverage: number, btcCollateral: number	Open leveraged perpetual position
close_perp	positionId: string	Close open position at current mark price
set_tpsl	positionId: string, tp: number null, sl: number null	Set/update take-profit and stop-loss levels
cancel_tpsl	orderId: string	Cancel a pending TP/SL order